INCH-POUND
MS27651G
19 October 2012
SUPERSEDING
MS27650F
17 August 1973

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, HERMETICALLY SEALED, 115 V 400Hz

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

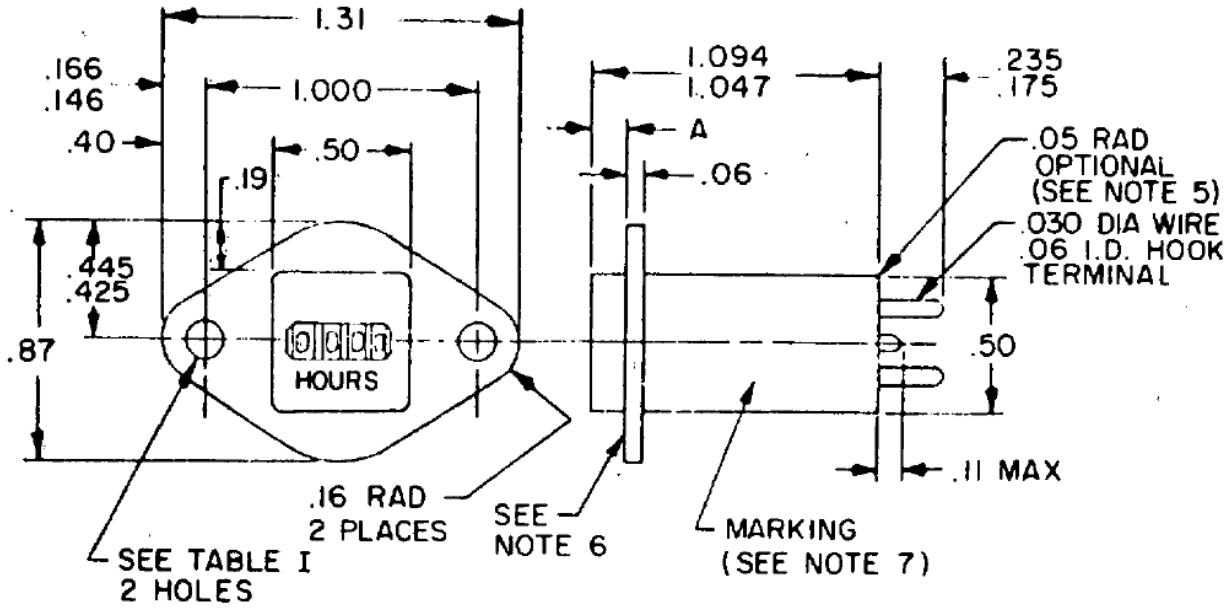


FIGURE 1. Meter configuration.

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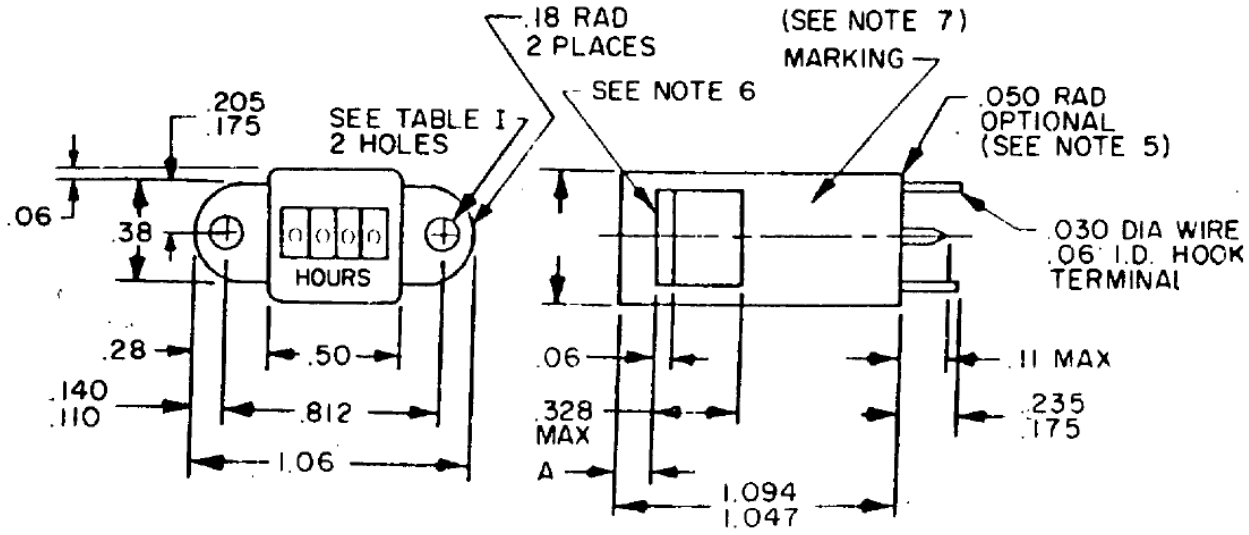


FIGURE 2. Alternate meter configuration.

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.02 for two place decimals and ±.005 for three place decimals
3. For design feature purposes, this specification sheet takes precedence over documents referenced herein.
4. Referenced documents shall be of the issue in effect on the date of invitation for bid.
5. Optional radius applies to all corners and edges.
6. Exposed tin plate on this surface of mounting flange only, per ASTM-B545, to provide an electrical ground for the case.
7. Marking surface is optional. Marking must not be obscured by mounting bracket.

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TABLE I. Mounting hole code letter.

Code Letter	Mounting Holes	Applicable Figure
A	NO. .112-40 UNC-2B Tapped Holes	1
B	NO. .112-40 UNC-2B Tapped Holes	1 (with flange rotated 90°)
C	.125 DIA Clearance Holes	1
D	.125 DIA Clearance Holes	1 (with flange rotated 90°)
E	NO. .112-40 UNC-2B Tapped Holes	2
F	NO. .112-40 UNC-2B Tapped Holes	2 (with flange rotated 90°)
G	.125 DIA Clearance Holes	2
H	.125 DIA Clearance Holes	2 (with flange rotated 90°)

REQUIREMENTS:

1. Dimensions. See figure 1 or figure 2, as applicable.

2. Electrical. Unless otherwise specified, all tests shall be at room pressure and temperature, approximately +25° C. All voltages shall be in accordance with MIL-DTL-7793, except lower voltage limit shall be 100 instead of 102.

3. Meter range. 0 to 10,000 hours, meter will read 9999 hours maximum.

4. Scale designator. Hours

5. Mounting flange. The surface of the mounting flange shall be tin plated per ASTM-B545, 0.0002 to 0.0004 inch thick to provide electrical ground for case (see figures 1 and 2).

6. Input power. The input power shall be 1.0 watt nominal, 1.5 watts maximum.

7. Input voltage. The input voltage shall be 115 volts ±15 volts.

8. Frequency. The frequency shall be 400 ±20 Hz.

9. Current. The current shall be 15 milliamperes maximum.

10. Accuracy. Accuracy shall be 0.1 percent or 1 hour, whichever is greater. Note: Accuracy is degraded at the same percent as the percentage change in input frequency plus the specified accuracy tolerance.

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11. Weight. The weight shall be a maximum of 0.7 ounce.

12. Transient protection. No temporary or permanent degradation or other malfunction shall be produced in the time totalizing meter if the input voltage should increase to +180 Vrms at 320 to 480 Hz for a period of 150 mlliseconds maximum.

13. Case finish. The case shall be finished in a durable, dull or semi-gloss black color.

14. Marking. The meter shall remain clearly and legibly marked with the following information after subjection to the environmental requirements:

(a) Part number, MS27651 (and dash number from table II and mounting hole code letter from table I). Example: MS27651-01.

(b) Source and date code per MIL-STD-1285.

(c) Nominal voltage and frequency.

TABLE II. Dash numbers and dimensions.

Dash Number	DIM. A ± .015	Dash Number	DIM. A ± .015
-01	.000	-14	.406
-02	.031	-15	.438
-03	.062	-16	.469
-04	.094	-17	.500
-05	.125	-18	.531
-06	.156	-19	.562
-07	.188	-20	.594
-08	.218	-21	.625
-09	.250	-22	.656
-10	.281	-23	.688
-11	.312	-24	.719
-12	.344	-25	.750*
-13	.375	-26	.781*

* Does not apply to figure 2.

15. Conformance inspection. The following tests, run in the order shown on all meters, supersede the individual tests and sampling Plans A and B specified in MIL-DTL-7793. The maximum meter reading at the time of delivery shall be 0001.

(a) Examination of product.

(b) Operational check. Ten operations for instant starting at 25° C ± 5° C. Five of these ten operations at 128 V, 380 Hz and the other five operations at 100V, 420 Hz.

(c) Thermal shock per MIL-STD-202, Method 107, test condition B, modified as follows:

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Test conditions:

1. The meter shall be operated at 128V, 380 Hz during Step 1.
2. The meter shall be operated at 100V, 420 Hz during Step 3.

MEASUREMENTS DURING CYCLING

1. The meter shall show no evidence of damage and shall not malfunction.

(d) Operation at room ambient temperature (approximately 25° C) and 115 V, 400 Hz for 20 ±1 hours.
(e) Hermetic seal.
(f) Insulation resistance.
(g) Sea level dielectric stress (600V RMS, 2 to 5 seconds)
(h) All meters failing one or more of these conformance tests shall be rejected.

16. In addition to the above tests, three sample meters total (any part number) shall be subjected to the following tests every two (2) years.

(a) Salt spray.
(b) Vibration.
(c) Shock.
(d) Life.

(1) High voltage at high temperature	(100 Hours)
(2) Low voltage at high temperature	(100 Hours)
(3) High voltage at low temperature	(100 Hours)
(4) Low voltage at low temperature	(100 Hours)
(5) 115 Volts at room ambient. (approximately 25º C)	(100 Hours)
	500 Hours

At the conclusion of these tests, meter readings shall be within the accuracy specified in Table I.

Qualification test.

The qualification tests are to be conducted in accordance with MIL-DTL-7793, except that four (4) sample meters (designated 1, 2, 3 and 4) shall be submitted to Group 1 tests per table III of this specification sheet. Samples designated 1 and 2 shall be submitted to Group 2 tests and samples designated 3 and 4 shall be submitted to Group 3 tests.

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TABLE III. Qualification tests.

Test	Requirement Paragraph	Test Paragraph
Group 1		
Examination of product	3.8.1	Visual examination (4.4.1)
	3.7.2	
	3.7.3	
	3.7.6	
	3.9	
Operational check	3.8.2	4.4.2
Group 2		
Power supply tolerance	3.8.4	4.4.4
Insulation resistance	3.8.5	4.4.5
Dielectric strength	3.8.6	4.4.6
Altitude	3.8.7	4.4.7
Thermal shock	3.8.8	4.4.8
Moisture resistance	3.8.9	4.4.9
Salt spray	3.8.10	4.4.10
Vibration	3.8.11	4.4.11
Shock	3.8.12	4.4.12
Hermetic seal	3.8.13	4.4.13
Group 3		
Life accuracy	3.8.14	4.4.14
Electromagnetic compatibility	3.8.15	4.4.15

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6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents: In addition to MIL-DTL-7793, this specification sheet references MIL-STD-202, MIL-STD-1285, and ASTM-B545.

Custodians:
Navy - AS
Air Force - 99
DLA - GS1

Preparing activity:
Navy - AS
(Project 6645-2012-022)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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